



WATCHSHELF

ENTERTAINMENT HUB GRADUATION PROJECT

PRESENTED TO THE FACULTY OF COMPUTER SCIENCE • 2026

A Professional Technical Documentation and Software Proposal for a Modern
Full-Stack Video Streaming Ecosystem.

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01

EXECUTIVE

SUMMARY

PROJECT OVERVIEW

WatchShelf is a premium full-stack entertainment platform designed to bridge the gap between Movies, TV Shows, and Anime. Inspired by industry leaders like Netflix and IMDb, it provides a centralized hub for discovery, streaming, and community interaction.

THE VISION

To create a performance-driven, responsive web application that caters to modern content consumption habits with a focus on ease of use and cinematic aesthetics.

THE CORE

A Django-powered backend providing a robust REST API, consumed by a lightning-fast React frontend optimized for dark-mode experiences.

02 TECHNOLOGY ARCHITECTURE

Our stack is selected for scalability, security, and developer productivity.

■ FRONTEND ENGINE

React.js with Vite for near-instant HMR and optimized production builds. Tailwind CSS for the cinematic dark-themed UI components.

React 18

Vite

Axios

Redux Toolkit

■ BACKEND CORE

Django REST Framework (DRF) handling complex relational data, JWT authentication, and media management.

Python 3.12

Django REST

JWT Auth

PostgreSQL

03

USER

ECOSYSTEM

ACCESS CONTROL LEVELS

SUPERUSER

The system architect with full governance over user memberships, platform configurations, and global analytics.

MANAGER

Content curators responsible for the library, genre categorization, and community moderation (comments/reviews).

END USER

The consumer. Access to discovery, search, watch history, ratings, and personalized "Shelf" favorites.

PERMISSION MATRIX

Feature	User	Manager	Super
Watch Content	✓	✓	✓
Rate/Comment	✓	✓	✓
CMS Access	-	✓	✓
User Management	-	-	✓
System Logs	-	-	✓

05 DATABASE & SCHEMA

A relational PostgreSQL architecture ensures data integrity and complex querying for the trending algorithm.

[ENTITY RELATIONSHIP DIAGRAM PLACEHOLDER]

Users ↔ Roles | Movies/Series ↔ Genres | Series ↔ Seasons ↔ Episodes | Users ↔ History/
Favorites

Core Entities

CONTENT

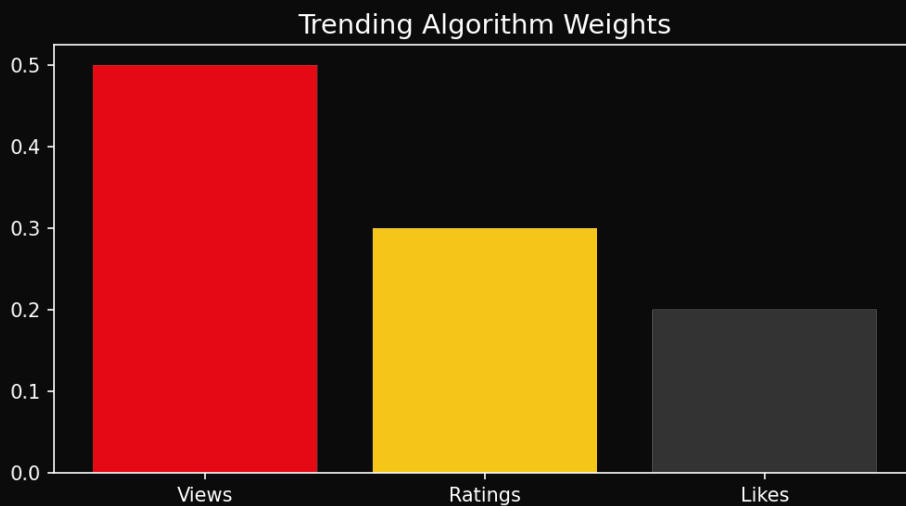
Universal metadata: Poster, Trailer, Rating, WatchCount.

INTERACTION

JWT Session linked comments, stars, and likes.

07 SMART TRENDING ANALYSIS

WatchShelf employs a weighted ranking algorithm to ensure the "Trending" page stays fresh and relevant.



$$\text{Score} = (\text{Total_Views} * 0.5) + (\text{Average_Rating} * 0.3) + (\text{Likes} * 0.2)$$

"Redefining the way we archive and experience digital entertainment."

WatchShelf stands as a testament to modern full-stack development, combining aesthetic excellence with technical rigor.

WATCHSHELF

DOUCMENT VERSION 1.0.0